

# Experiment 2

## Designing and Implementing ETL Processes using Microsoft Power BI

### AIM:

To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

### OBJECTIVES:

- To extract data from real-world business datasets using Microsoft Power BI.
- To transform and clean the extracted data using Power Query Editor.
- To apply data transformation techniques such as filtering, merging, removing duplicates, handling missing values, and changing data types
- To load the transformed data into a unified analytical model.

### TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Sample datasets: Global Superstore 2018.xlsx , Employee\_DB.xlsx and a sample SQL database

**DATASET USED:** Global Superstore 2018.xlsx ,Employee\_DB.xlsx , Transactions.sql

### THEORY:

1. **ETL** (Extract, Transform, Load) is one of the most important processes in Business Intelligence. It enables organizations to collect data from source systems, transform it into a clean and consistent format, and load it into a centralized environment for analysis.
2. **Extract Phase:** The extraction phase involves collecting data from source systems. In this experiment, data is extracted from the Orders, Returns, and People worksheets available in the Global Superstore dataset.
3. **Transform Phase:** The transformation phase includes data cleaning, handling missing values, correcting data types, creating derived columns, filtering invalid records, and integrating multiple datasets.
4. **Load Phase:** The transformed data is loaded into the Power BI data model where relationships can be established and analytical measures can be created using DAX.
5. **Power Query Editor** serves as the ETL engine in Power BI and records each transformation step as an Applied Step, making the workflow transparent and reproducible.

### PROCEDURE:

#### Part A: Extract-Connect to Dataset

**Step 1:** Download the Global Superstore Dataset from Kaggle.

+

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192

Code

Download

Global Super Store Dataset

Consumer and product Analytics on Global Super Store Data.

Data Card

Code (25)

Discussion (4)

Suggestions (0)

About Dataset

Context

Shopping online is currently the need of the hour. Because of this COVID, it's not easy to walk in a store randomly and buy anything you want. I this I am trying to understand a few things like

- Customers Analysis
  - Profile the customers based on their frequency of purchase - calculate frequency of purchase for each customer
  - Do the high frequent customers are contributing more revenue
  - Are they also profitable - what is the profit margin across the buckets
  - Which customer segment is most profitable in each year.
  - How the customers are distributed across the countries- -
- Product Analysis
  - Which country has top sales?
  - Which are the top 5 profit-making product types on a yearly basis
  - How is the product price varying with sales - Is there any increase in sales with the decrease in price at a day level

Usability

8.24

License

Database: Open Database, Conte...

Expected update frequency

Monthly

Tags

Business

Online Communities

Retail and Shopping

E-Commerce Services

## Step 2: Open Power BI Desktop.

Home

Open

Sign in

Options and settings

About

Select a data source or start with a blank report

Blank report

OneLake catalog

Excel workbook

SQL Server

Learn with sample data

Get data from other sources

Recommended

Getting started

Intro—What is Power BI?

Recent

Shared with me

Filter by keyword

Filter

Name

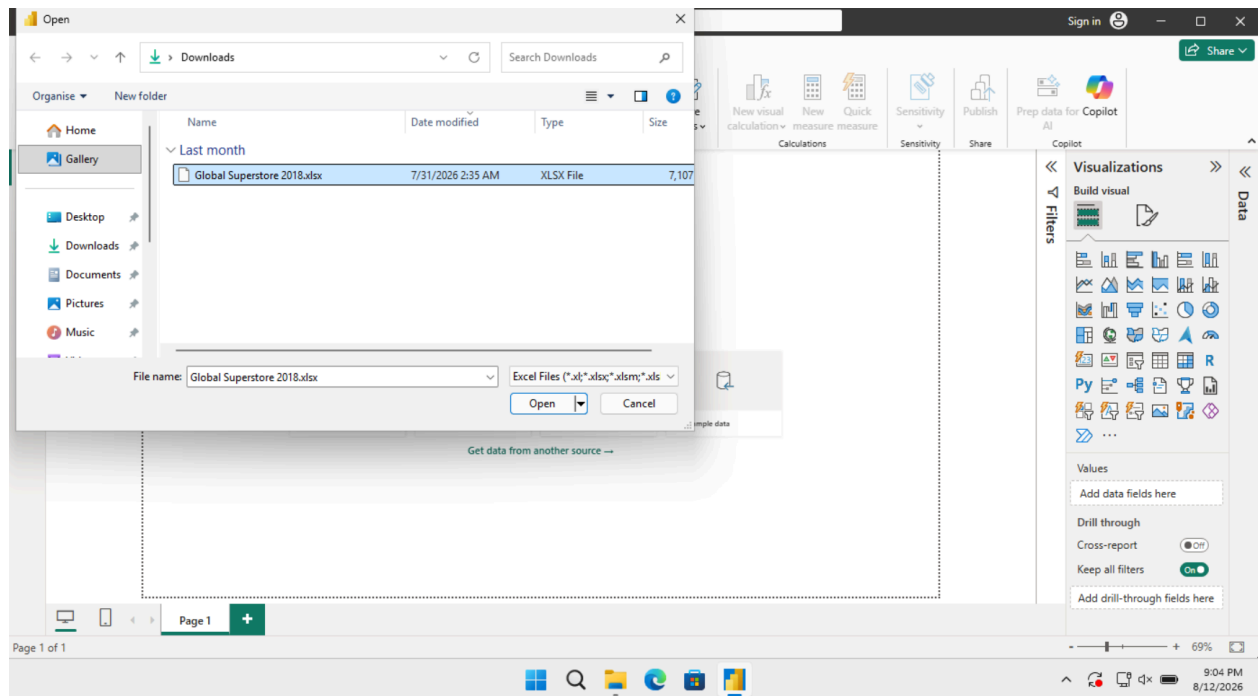
Location

Opened

Windows taskbar

9:03 PM 8/12/2026

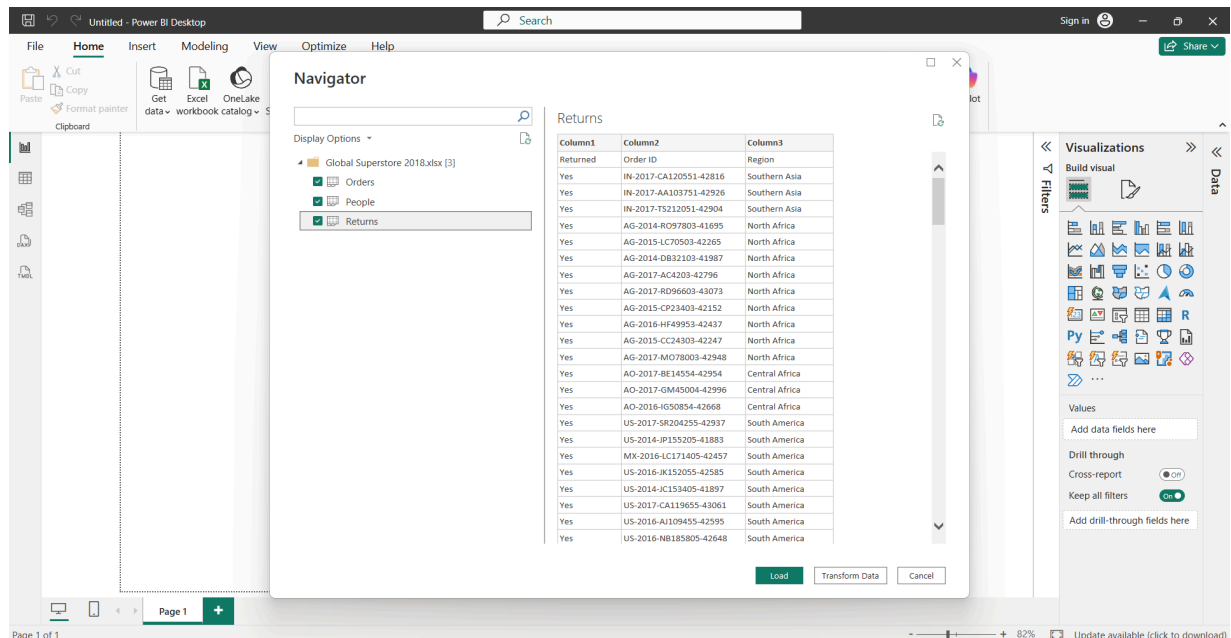
Home > Get Data > Excel Workbook  
 Select Global Superstore Dataset.xlsx



Click Transform Data.

**Step 3:** Import the following worksheets:

- Orders
- Returns
- People



Verify that all imported tables appear in the Queries pane.

**Part B: Transform-Power Query Editor**

**Step 4:** Open Power Query Editor.

FileHomeTransformAdd ColumnViewToolsHelp

Close & Apply

New Source

Recent Sources

Enter Data

Data source settings

Manage Parameters

Export query results

Refresh Preview

Properties

Advanced Editor

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Sort

Split Column

Group By

Data Type: Whole Number

Use First Row as Headers

Replace Values

Merge Queries

Append Queries

Combine Files

Combine

Queries [3]

fx

= Table.SelectRows(#"Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(\_), {"", ""})))

Query Settings

PROPERTIES

Name

Orders

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Removed Duplicates

Removed Blank Rows

| Row ID | Order ID | Order Date             | Ship Date  | Ship Mode  | Customer ID    |           |
|--------|----------|------------------------|------------|------------|----------------|-----------|
| 1      | 24599    | IN-2017-CA120551-42816 | 3/22/2017  | 3/29/2017  | Standard Class | CA-120551 |
| 2      | 29465    | ID-2015-BD116051-42248 | 9/1/2015   | 9/4/2015   | Second Class   | BD-116051 |
| 3      | 24598    | IN-2017-CA120551-42816 | 3/22/2017  | 3/29/2017  | Standard Class | CA-120551 |
| 4      | 24597    | IN-2017-CA120551-42816 | 3/22/2017  | 3/29/2017  | Standard Class | CA-120551 |
| 5      | 29464    | ID-2015-BD116051-42248 | 9/1/2015   | 9/4/2015   | Second Class   | BD-116051 |
| 6      | 28879    | ID-2015-AJ107801-42113 | 4/19/2015  | 4/22/2015  | First Class    | AJ-107801 |
| 7      | 27993    | IN-2017-GM144551-42948 | 8/1/2017   | 8/5/2017   | Standard Class | GM-144551 |
| 8      | 28967    | IN-2017-VB217451-43080 | 12/11/2017 | 12/15/2017 | Standard Class | VB-217451 |
| 9      | 29492    | IN-2016-LO171701-42637 | 9/24/2016  | 9/28/2016  | Standard Class | LO-171701 |
| 10     | 28966    | IN-2017-VB217451-43080 | 12/11/2017 | 12/15/2017 | Standard Class | VB-217451 |
| 11     | 25232    | ID-2015-SS201401-42354 | 12/16/2015 | 12/20/2015 | Standard Class | SS-201401 |
| 12     | 23222    | IN-2017-AA103751-42926 | 7/10/2017  | 7/15/2017  | Second Class   | AA-103751 |
| 13     | 29094    | IN-2015-BG110351-42275 | 9/28/2015  | 10/4/2015  | Standard Class | BG-110351 |
| 14     | 28265    | IN-2016-AH105851-42701 | 11/27/2016 | 12/1/2016  | Standard Class | AH-105851 |
| 15     | 27278    | IN-2016-CS118451-42387 | 1/18/2016  | 1/20/2016  | First Class    | CS-118451 |
| 16     | 27279    | IN-2016-CS118451-42387 | 1/18/2016  | 1/20/2016  | First Class    | CS-118451 |
| 17     | 29096    | IN-2015-BG110351-42275 | 9/28/2015  | 10/4/2015  | Standard Class | BG-110351 |
| 18     | 23219    | IN-2017-AA103751-42926 | 7/10/2017  | 7/15/2017  | Second Class   | AA-103751 |
| 19     | 28276    | IN-2014-AH105851-41973 | 11/30/2014 | 12/3/2014  | First Class    | AH-105851 |
| 20     | 29585    | IN-2015-DW131951-42160 | 6/5/2015   | 6/10/2015  | Standard Class | DW-131951 |
| 21     | 23951    | IN-2014-RS194201-41867 | 8/16/2014  | 8/18/2014  | First Class    | RS-194201 |
| 22     | 24892    | IN-2017-TS213401-43092 | 12/23/2017 | 12/28/2017 | Standard Class | TS-213401 |
| 23     | 20728    | IN-2017-JG151151-43032 | 10/24/2017 | 10/27/2017 | First Class    | JG-151151 |

24 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

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**Step 5:** Apply the following transformations to the Orders table:

- Remove duplicate records
- Remove blank rows
- Verify and correct data types
- Rename columns where required
- Extract Year and Month from Order Date
- Create a custom Profit Margin column
- Standardize Region and Segment values

FileHomeTransformAdd ColumnViewToolsHelp

Close & Apply

New Source

Recent Sources

Enter Data

Data source settings

Data Sources

Manage Parameters

Export query results

Refresh Preview

Advanced Editor

Properties

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Sort

Split Column

Group By

Data Type: Whole Number

Use First Row as Headers

Replace Values

Merge Queries

Append Queries

Combine Files

Combine

Queries [3]

Orders

People

Returns

fx

= Table.SelectRows(#"Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(\_), {"", ""})))

123 Row ID

Order ID

Order Date

Ship Date

Ship Mode

Customer ID

1

24599

IN-2017-CA120551-42816

3/22/2017

3/29/2017

Standard Class

CA-120551

2

29465

ID-2015-BD116051-42248

9/1/2015

9/4/2015

Second Class

BD-116051

3

24598

IN-2017-CA120551-42816

3/22/2017

3/29/2017

Standard Class

CA-120551

4

24597

IN-2017-CA120551-42816

3/22/2017

3/29/2017

Standard Class

CA-120551

5

29464

ID-2015-BD116051-42248

9/1/2015

9/4/2015

Second Class

BD-116051

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28879

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9/28/2016

Standard Class

LO-171701

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12/11/2017

12/15/2017

Standard Class

VB-217451

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12/16/2015

12/20/2015

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8/16/2014

8/18/2014

First Class

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12/23/2017

12/28/2017

Standard Class

TS-213401

23

20728

IN-2017-JG151151-43032

10/24/2017

10/27/2017

First Class

JG-151151

24

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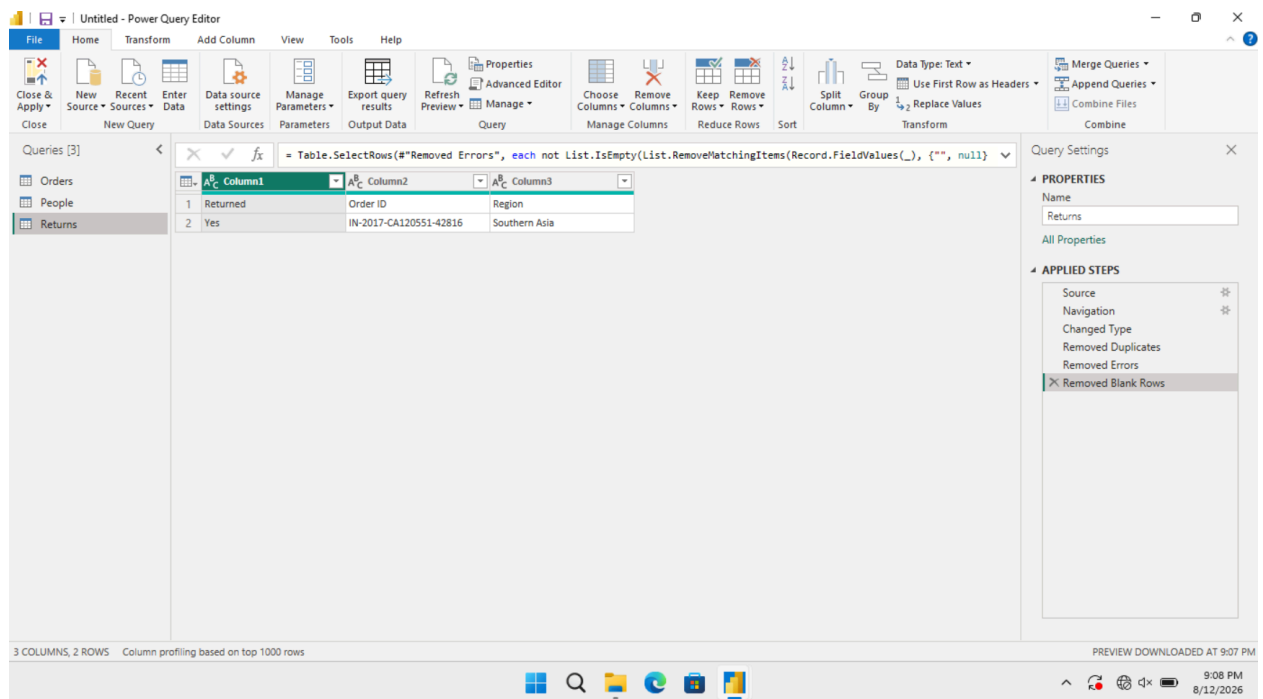
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8/12/2026

**Step 6:** Transform the Returns table:

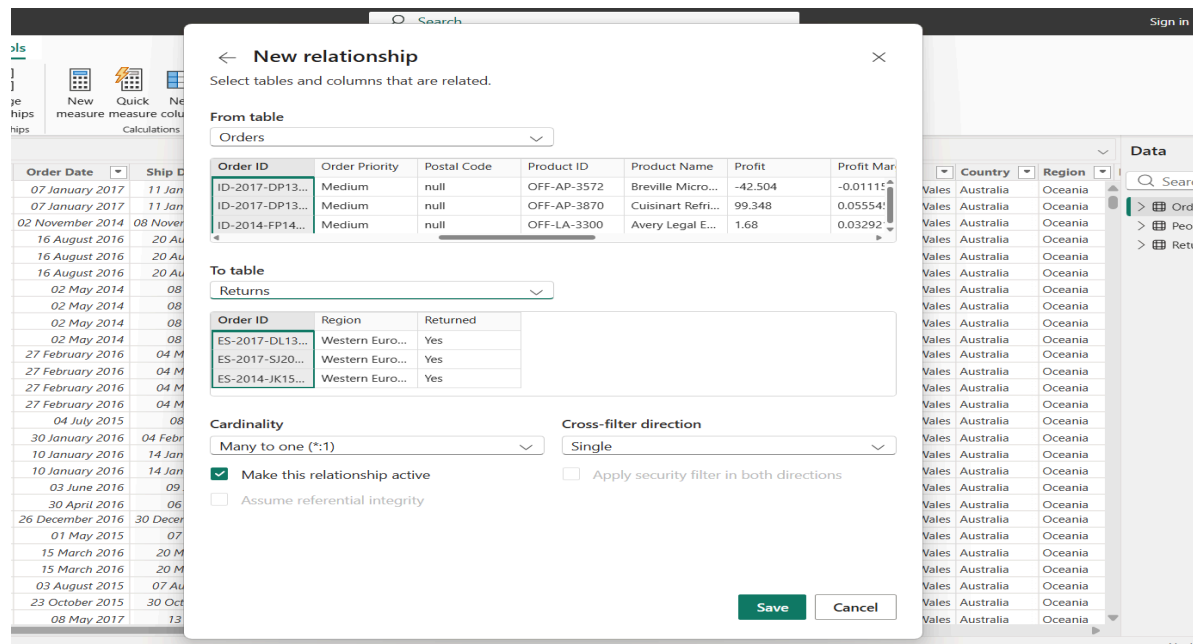
- Remove duplicate Order IDs
- Check for missing values
- Standardize Returned field values



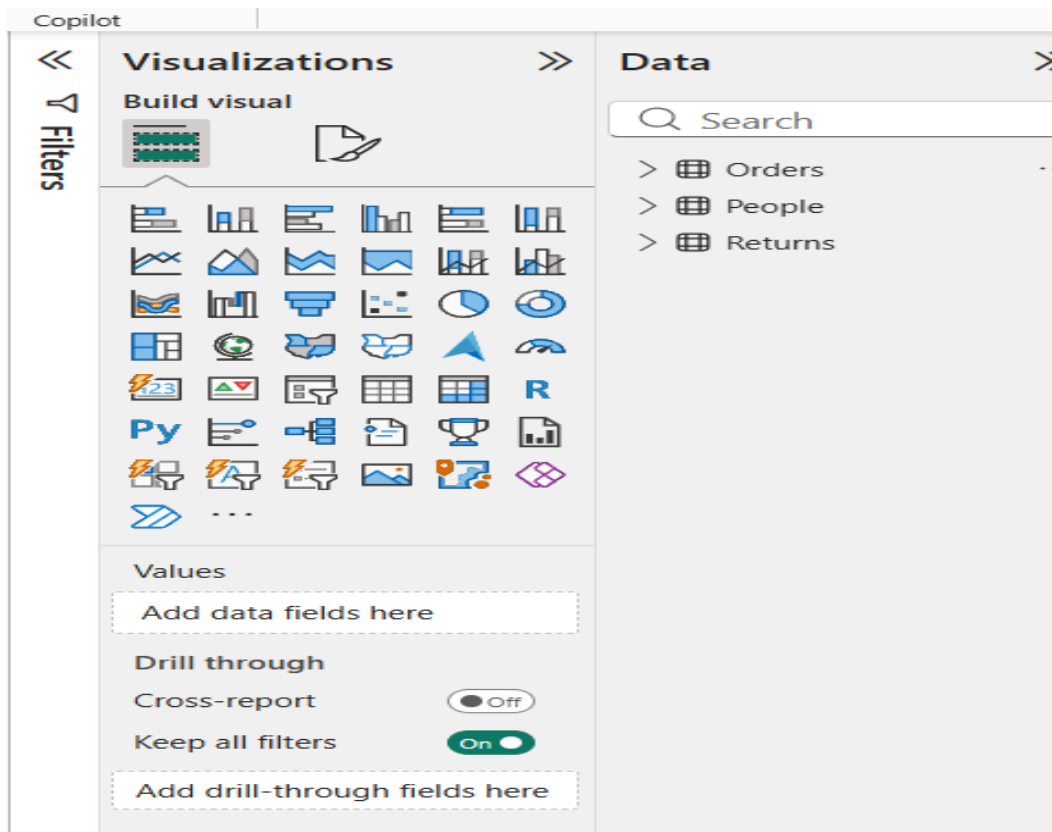
**Step 7:** Transform the People table:

- Validate regional assignments
- Remove unnecessary columns
- Verify consistency of region names

**Step 8:** Merge Orders and Returns tables using Order ID.

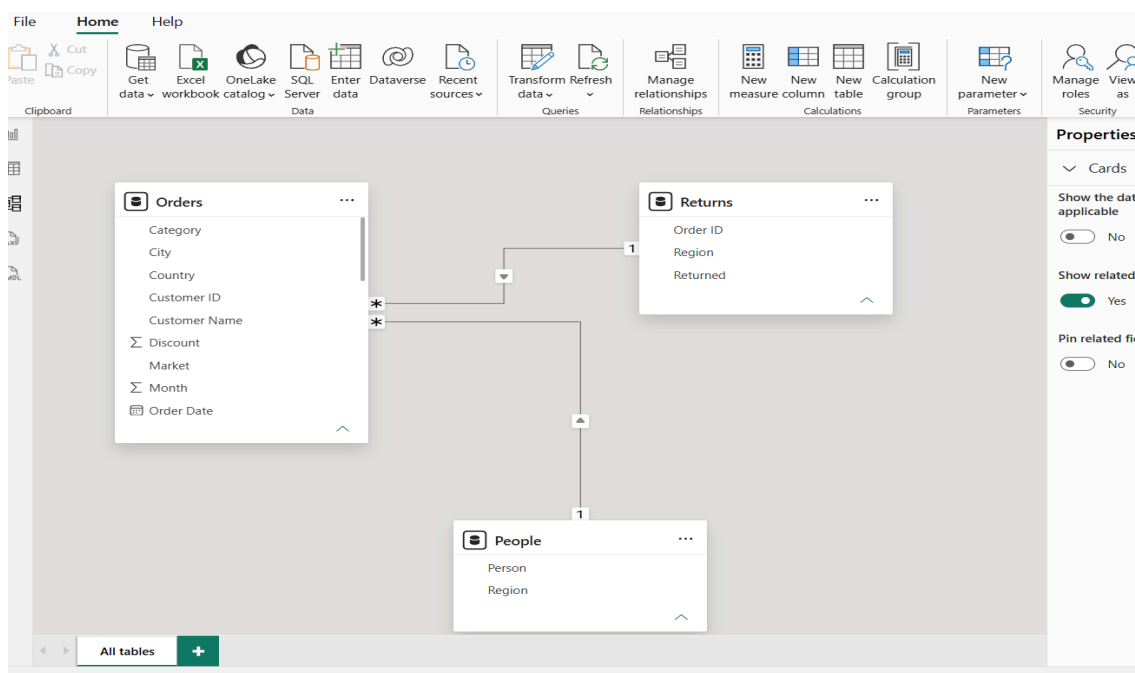


**Step 9:** Review all Applied Steps and click Close & Apply.



## Part C: Load-Build Data Model

**Step 10:** Navigate to Model View.



**Step 11:** Create relationships:

Orders ↔ Returns (Order ID)

Orders ↔ People (Region)

## Manage relationships



| <a href="#">+ New relationship</a>                         | <a href="#">Autodetect</a> | <a href="#">Edit</a> | <a href="#">Delete</a> | <a href="#">Filter</a> v |
|------------------------------------------------------------|----------------------------|----------------------|------------------------|--------------------------|
| <input checked="" type="checkbox"/> From: table (column) ↑ | Relationship               | To: table (column)   | Status                 |                          |
| <input checked="" type="checkbox"/> Orders (Order ID)      | * — 1                      | Returns (Order ID)   | Active                 | ...                      |
| <input checked="" type="checkbox"/> Orders (Region)        | * — 1                      | People (Region)      | Active                 | ...                      |

### Step 12: Create DAX measures:

Total Sales = SUM(Orders[Sales])

Total Profit = SUM(Orders[Profit])

Profit Margin = DIVIDE(SUM(Orders[Profit]),SUM(Orders[Sales]))

Total Orders = DISTINCTCOUNT(Orders[Order ID])

### Step 13: Create a simple report visual using fields from related tables.



## RESULT:

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

## LEARNING OUTCOMES:

- Explain the ETL process and its role in Business Intelligence systems.
- Extract data from real-world business datasets using Power BI.
- Apply transformation techniques including cleaning, filtering, and merging.
- Create relationships and analytical measures in Power BI

- Validate transformed datasets for reporting and decision-making.